

Spec to RTL

Before you write bit-level logic, translate the spec into structure

UART TX starter

- Starter preset: UART transmit eight-N-one
- Tx busy, plus baud parameter BAUD_DIV
- Required blocks still waiting
- Check the block rows, then click Generate skeleton to grow a uart_tx module outline

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Spec → RTL checklist

Turn a one-page protocol spec into an RTL **skeleton**: ports, parameters, block list, then stub modules — before you write bit-level logic. Starter: **UART TX** with clk/rst and bus ports checked.

Starter example: UART TX spec with clock/reset, bus ports, and BAUD_DIV already checked. Add blocks and click *Generate skeleton*.

Load starter example

Challenges 0 / 22 cleared

Quiz: order: Spec → RTL usually starts with...

☐ ports and block diagram from the spec

☐ random testbench first

☐ synthesis constraints only

☐ VIP agents before ports

Show hint

Check

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Idle

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Core ideas

Ports first

clk/rst + bus-facing pins before logic.

Parameters

Baud, mode, widths from the spec.

Blocks

Name generators, shifters, FSMs on paper.

Skeleton

Stub module + TODO — not finished RTL.

Spec to RTL starter

learn_uart — spec to rtl

Real RTL/TB practice

- In Track A, from the UART TX spec list every port name and one parameter
- Name three blocks you would draw on paper before coding
- Optional: compare your list to a UART sketch in the legacy combined materials
- Write one sentence on what belongs in the skeleton versus what belongs in the FSM

Pitfalls to watch

- Do not skip ports and jump straight to the FSM
- Parameters are not optional when baud or mode defines behavior
- A skeleton with TODO is not done RTL, this lab separates planning from implementation
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish required UART checks and generate the skeleton once
- On paper, list ports, parameters, and three blocks for UART TX
- When you are ready, take the short quiz, then continue to baud divider